Internal Consistency Verification of the Muon Shower Coincidence Analysis Pipeline

Muon Shower Detector Array — Technical Note

March 2026

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1 Overview and Scope

This document presents an independent verification of the internal consistency of three coupled components of the muon shower coincidence analysis:

1. `find_coincidences()` — the greedy one-sided sliding-window algorithm that identifies multi-detector coincidence clusters.
2. `expected_accidentals()` — the analytic formula $E_k = T_{\text{eff}}\,(2\tau)^{k-1} \sum_{|S|=k} \prod_{i\in S} r_i$ for the expected number of accidental k-fold coincidences.
3. `significance()` — the one-sided Poisson tail probability converted to Gaussian standard deviations.

The verification works through the mathematics from first principles, rather than relying solely on code reading. The central finding is that the algorithm and formula are **exactly consistent for** $k = 2$ but exhibit a **conservative mismatch for** $k \geq 3$: the formula overestimates the expected accidentals relative to what the algorithm actually counts, resulting in a slight underestimate of the reported statistical significance.

2 Notation and Setup

Definition 1 (Detector rates and run parameters)**.** Let $N = 8$ detectors fire independently as stationary Poisson processes with rates $r_1, \ldots, r_8$ (Hz). The effective run duration (after sync and jump vetoes) is T_{eff} (s).

Definition 2 (Algorithm window)**.** The greedy algorithm parameter `window_us` defines a one-sided search window of width W (in microseconds, converted to seconds as needed). For each seed event at time t_0, all unused events from different detectors in the interval $[t_0,\, t_0 + W]$ are collected into a cluster.

Definition 3 (Formula parameter)**.** The accidental formula uses a parameter τ set equal to W in the code:

$$\tau \equiv W, \qquad 2\tau = 2W. \tag{1}$$

The formula for the expected number of k-fold accidentals is

$$E_k = T_{\text{eff}}\,(2\tau)^{k-1} \sum_{\substack{S\subseteq\{1,\ldots,8\}\\ |S|=k}} \prod_{i\in S} r_i\,. \tag{2}$$

3 Consistency Analysis for $k = 2$

3.1 What the greedy algorithm counts

All calibrated events are sorted by corrected timestamp. The algorithm walks through them in order. When it reaches an unused event from detector i at time t_0, it searches $[t_0,\, t_0 + W]$ for an unused event from a different detector j.

Consider an arbitrary pair of events: event A from detector i at time t_A and event B from detector j at time t_B, with $t_A < t_B$ and $t_B - t_A < W$. When the algorithm reaches A (which is earlier), it opens the window $[t_A,\, t_A + W]$. Since $t_B \in [t_A,\, t_A + W]$, the pair is found.

Conversely, if $t_B < t_A$ and $t_A - t_B < W$, then B is the seed and A is found in $[t_B,\, t_B + W]$.

Claim 1. *The greedy algorithm finds every pair of events from distinct detectors with $|\Delta t| < W$, regardless of which detector fires first.*

3.2 Expected count from the Poisson model

For a specific unordered pair $\{i, j\}$, the expected number of accidental pairs with $|\Delta t| < W$ is computed by decomposing into the two temporal orderings:

- Detector i fires first, j follows within W: $r_i\, T_{\text{eff}} \times r_j\, W$
- Detector j fires first, i follows within W: $r_j\, T_{\text{eff}} \times r_i\, W$

Total:

$$E^{\text{greedy}}_{2,\{i,j\}} = 2\, r_i\, r_j\, W\, T_{\text{eff}}. \tag{3}$$

3.3 Comparison with the formula

Setting $k = 2$ and $\tau = W$ in Eq. (2):

$$E^{\text{formula}}_{2,\{i,j\}} = T_{\text{eff}}\, (2W)^{2-1}\, r_i\, r_j = 2\, r_i\, r_j\, W\, T_{\text{eff}}. \tag{4}$$

$$\boxed{E^{\text{greedy}}_2 = E^{\text{formula}}_2 \quad \checkmark} \tag{5}$$

The algorithm and formula are **exactly consistent** for $k = 2$.

4 Consistency Analysis for $k = 3$

4.1 What the greedy algorithm counts

For a specific unordered triple $\{a, b, c\}$, the algorithm finds the cluster when the earliest of the three events (the seed) has both other events within $[t_{\min},\, t_{\min} + W]$. The condition is:

$$\text{span} \;\equiv\; t_{\max} - t_{\min} \;<\; W. \tag{6}$$

Decomposing by which detector fires earliest (three orderings for an unordered triple):

$$E^{\text{greedy}}_{3,\{a,b,c\}} = 3\, r_a\, r_b\, r_c\, W^2\, T_{\text{eff}}. \tag{7}$$

Here each term contributes $r_{\text{seed}}\, T_{\text{eff}} \times (r_{\text{follower}_1}\, W)(r_{\text{follower}_2}\, W)$, and there are 3 choices of seed (one for each detector being the earliest).

4.2 What the formula predicts

Setting $k = 3$ and $\tau = W$ in Eq. (2):

$$E^{\text{formula}}_{3,\{a,b,c\}} = T_{\text{eff}}\, (2W)^2\, r_a\, r_b\, r_c = 4\, r_a\, r_b\, r_c\, W^2\, T_{\text{eff}}. \tag{8}$$

4.3 Identifying the discrepancy

Comparing Eqs. (7) and (8):

$$\frac{E^{\text{formula}}_3}{E^{\text{greedy}}_3} = \frac{4}{3} \approx 1.33. \tag{9}$$

The formula overestimates the expected accidentals by a factor of 4/3 relative to what the greedy algorithm actually counts.

4.4 Origin of the mismatch

The formula is derived using a **symmetric window** of half-width τ centered on a designated trigger detector. With detector a as the trigger at time t, the condition is that both b and c fire in $[t-\tau,\, t+\tau]$. This allows configurations where one follower is *before* the trigger and one is *after*, giving a span up to $2\tau = 2W$.

The greedy algorithm, by contrast, seeds on the *earliest* event and searches only *forward*. The effective span limit is W, not $2W$.

4.5 Geometric calculation of the mismatch

Consider the symmetric-window framework: the trigger fires at time 0, and two followers t_b, t_c are independently uniform on $[-\tau, \tau]$ (in the thin-window limit). What fraction of these triples have span $\leq \tau = W$?

Let $u = t_b/\tau$ and $v = t_c/\tau$, both uniform on $[-1, 1]$. The span is $\text{span}/\tau = \max(0, u, v) - \min(0, u, v)$.

Case 1: $u \geq 0,\ v \geq 0$ (probability 1/4). Span $= \max(u, v) \leq 1$. Fraction with span > 1: 0.

Case 2: $u \leq 0,\ v \leq 0$ (probability 1/4). Span $= -\min(u, v) \leq 1$. Fraction with span > 1: 0.

Case 3: $u > 0,\ v < 0$ (probability 1/4). Span $= u + |v| = u - v$. Probability that $u - v > 1$ where $u \in [0, 1]$, $v \in [-1, 0]$: set $v' = -v \in [0, 1]$; then $P(u + v' > 1) = \frac{1}{2}$ (area of the triangle above the diagonal in the unit square). Contribution: $(1/4)(1/2) = 1/8$.

Case 4: $u < 0,\ v > 0$ (probability 1/4). By symmetry with Case 3: contribution 1/8.

Total:

$$P(\text{span} > \tau) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}, \qquad P(\text{span} \leq \tau) = \frac{3}{4}. \tag{10}$$

The greedy algorithm captures the 3/4 fraction. Since the formula counts all triples (including the 1/4 with span in $(\tau, 2\tau]$), the ratio is:

$$\frac{E_3^{\text{formula}}}{E_3^{\text{greedy}}} = \frac{1}{3/4} = \frac{4}{3} \quad \checkmark \tag{11}$$

confirming the direct calculation.

5 Generalisation to Arbitrary k

For a k-fold coincidence, the formula uses $k-1$ followers each uniform on $[-\tau, \tau]$. The greedy algorithm requires the span (max minus min of all k events, including the trigger at 0) to be at most τ.

The overcounting factor is:

$$R_k = \frac{E_k^{\text{formula}}}{E_k^{\text{greedy}}} = \frac{1}{P(\text{span} \leq \tau)}, \tag{12}$$

where the span is computed with $k-1$ points uniform on $[-\tau, \tau]$ plus the trigger at 0.

For general k, the expected number of triples with all k events in a window of width S, when one event (the trigger) is fixed and the other $k-1$ are independent, can be decomposed by which of the k events is earliest. When the trigger is at the boundary (as in the greedy algorithm), only 1 out of k orderings has the trigger as the earliest. By symmetry of the uniform distribution, each of the k positions (trigger plus $k-1$ followers) is equally likely to be the minimum.

A more direct route: the greedy algorithm counts triples with span $< W$, while the formula effectively counts triples with all $k-1$ followers within $\pm W$ of the trigger. The expected count from the formula (for a fixed trigger, one-directional) is:

$$E_k^{\text{formula, trigger } a} = r_a\, T_{\text{eff}} \prod_{m=2}^{k} (r_{i_m} \cdot 2W). \tag{13}$$

The greedy expected count, decomposed by which detector is earliest, sums k terms each contributing $(r_{\text{seed}}\, T_{\text{eff}}) \prod (r_j \cdot W)$:

$$E_k^{\text{greedy}} = k\, W^{k-1}\, T_{\text{eff}} \prod_{i \in S} r_i. \tag{14}$$

The ratio is:

$$R_k = \frac{(2W)^{k-1}}{k\, W^{k-1}} = \frac{2^{k-1}}{k}. \tag{15}$$

k	$R_k = 2^{k-1}/k$	Formula overestimate
2	$2/2 = 1$	None (exact)
3	$4/3 \approx 1.33$	33%
4	$8/4 = 2$	100%
5	$16/5 = 3.2$	220%

Table 1: Ratio of the formula's prediction to the greedy algorithm's actual expected count, by fold order.

Remark. The overcounting grows with k, but the absolute expected counts E_k shrink dramatically with k (by a factor of $\sim r \cdot 2\tau \approx 10^{-3}$ per additional fold). For $k \geq 4$, E_k is so small that the overcounting has negligible practical impact on the significance.

6 Impact on Reported Significance

Because the formula overestimates E_k for $k \geq 3$, the reported significance is a **lower bound** (conservative). The corrected expected accidentals and approximate significance shifts are:

Channel	E_k^{formula}	R_k	$E_k^{\text{corrected}}$	σ_{reported}	$\sigma_{\text{corrected}}$
2-fold, 1 ms	202	1.00	202	$+4.8\sigma$	$+4.8\sigma$
3-fold, 1 ms	0.80	1.33	0.60	$+4.1\sigma$	$\sim+4.4\sigma$
5-fold, 1 ms	3×10^{-4}	3.2	9×10^{-5}	$+4.5\sigma$	$\sim+4.7\sigma$

Table 2: Effect of the overcounting correction on the key results. Changes are modest and always increase the significance.

The strongest statistical channel (2-fold at 1 ms, $+4.8\sigma$) is **completely unaffected**.

7 Verification of `significance()`

7.1 The p-value calculation

The code computes:

$$p = 1 - F_{\text{Poisson}}(O - 1;\, E_k), \tag{16}$$

where $F_{\text{Poisson}}(n;\lambda) = P(X \le n)$ for $X \sim \text{Poisson}(\lambda)$.

Expanding:

$$p = 1 - P(X \le O - 1) = P(X \ge O). \tag{17}$$

This is the correct one-sided upper-tail probability: the probability of observing O *or more* events under the null hypothesis. This matches Eq. (7.1) in the Statistical Significance derivation document.

7.2 Off-by-one verification

The critical question is whether the code computes $P(X \ge O)$ or $P(X > O) = P(X \ge O + 1)$.

- `poisson.cdf(n, λ)` in SciPy returns $P(X \le n)$.
- The code calls `poisson.cdf(observed - 1, expected)`, obtaining $P(X \le O - 1)$.
- Then $p = 1 - P(X \le O - 1) = P(X \ge O)$. ✓

Had the code mistakenly used `poisson.cdf(observed, expected)`, it would compute $P(X \ge O + 1)$, which excludes the observed value itself and *understates* the significance. The code is correct.

7.3 Gaussian conversion

The code computes:

$$\sigma = \Phi^{-1}(1 - p), \tag{18}$$

where Φ^{-1} is the inverse CDF (percent-point function) of the standard normal distribution. This finds the z-score whose upper-tail area equals p:

$$P(Z > \sigma) = p, \qquad Z \sim \mathcal{N}(0, 1). \tag{19}$$

This is the standard one-sided conversion used in particle physics. The sign convention is correct: $\sigma > 0$ for excesses ($O > E_k$), $\sigma < 0$ for deficits. ✓

7.4 Edge cases

1. $O = 0$**:** `poisson.cdf(-1,λ)` returns 0 in SciPy, so $p = 1$. Then $\sigma = \Phi^{-1}(0) = -\infty$. Mathematically correct: observing zero events is maximally unsurprising in the upward direction.

2. $E_k = 0$**:** The function returns `nan`, which is appropriate — a Poisson distribution with $\lambda = 0$ is degenerate at $X = 0$, and a hypothesis test against it is undefined.

3. **Very large excess:** The floor $p \leftarrow \max(p, 10^{-15})$ caps σ at approximately 7.94, preventing floating-point underflow in the `ppf` call. For the current data ($\sigma_{\max} \approx 4.8$), this floor is never reached.

8 Additional Systematic Considerations

8.1 Greedy event consumption

The greedy algorithm consumes each event into at most one cluster. If a genuine shower event happens to fall within W of an unrelated earlier event, it may be absorbed into an accidental 2-fold pair, removing it from the pool and potentially preventing a higher-fold cluster from forming.

The probability of this occurring for any given shower event is of order $r\tau \sim 10^{-3}$ per event. For a 5-fold shower, the probability that *any* of its 5 constituent events are pre-consumed is $\sim 5 \times 10^{-3}$. This is negligible and biases the result in the conservative direction (undercounting genuine high-fold coincidences).

8.2 One hit per Nano per cluster

The algorithm enforces at most one event per detector per cluster via the `seen_nanos` dictionary. The formula also implicitly assumes this: each detector contributes exactly one factor of r_i to the rate product. These are consistent.

8.3 Calibration timing errors

The out-of-sample calibration RMS is $\sim$1.26 ms. At the 1 ms window, calibration errors are comparable to the window width. This has two competing effects:

- Genuine coincidences may be smeared outside the window, reducing the observed count (conservative bias).
- Accidental near-pairs may be shifted into coincidence, increasing the observed count.

For tight genuine clusters (the 5-fold candidate spans only 3.6 μs), even a 1 ms calibration error leaves the cluster well within the window. The accidental formula does not account for calibration noise, but since the errors are symmetric, zero-mean, and small relative to inter-event spacing, the net bias is negligible.

8.4 Non-Poisson rate fluctuations

The formula assumes stationary Poisson processes. If detector rates fluctuate (e.g., due to temperature-dependent gain), Jensen's inequality implies $\langle r_i r_j \rangle \geq \langle r_i \rangle \langle r_j \rangle$ for positively correlated rate variations, making the true accidental rate higher than the formula predicts. The sync veto, jump veto, and null test (200 random time-shifts) collectively address this concern: the null test provides an analysis-independent check that the accidental baseline is correctly estimated.

9 Summary of Findings

Component	**Status**	**Notes**
Algorithm $\leftrightarrow$ Formula, $k = 2$	Exactly consistent	Both count pairs with $\lvert\Delta t\rvert < W$
Algorithm $\leftrightarrow$ Formula, $k \geq 3$	Conservative mismatch	Formula overestimates by $2^{k-1}/k$
`significance()` p-value	Correct	$P(X \geq O)$; no off-by-one
`significance()` σ conversion	Correct	One-sided $\Phi^{-1}(1-p)$
Greedy event consumption	Conservative bias	Negligible at $r\tau \sim 10^{-3}$
Calibration errors	Symmetric, small	Negligible net effect

Table 3: Summary of consistency verification results.

9.1 Principal finding

The only substantive finding is the mismatch between the symmetric-window accidental formula and the one-sided greedy algorithm for fold orders $k \geq 3$. The formula uses a symmetric window of half-width $\tau = W$ centered on a trigger, allowing event spans up to $2W$. The greedy algorithm seeds on the earliest event and searches forward to $t_0 + W$, limiting spans to at most W. The formula therefore overestimates the expected accidentals by a factor of $2^{k-1}/k$ (Table 1).

9.2 Practical impact

The mismatch is **conservative**: it inflates the expected background, making the reported significance a lower bound on the true significance. The strongest statistical channel ($k = 2$ at 1 ms, $+4.8\sigma$) is completely unaffected. The 3-fold significance would increase from $+4.1\sigma$ to approximately $+4.4\sigma$ under the corrected formula.

9.3 Possible resolutions

To achieve exact consistency between the formula and the algorithm, one could either:

1. **Modify the formula** to compute the expected number of k-tuples with span $< W$ (rather than all events within $\pm\tau$ of a trigger). The corrected formula is

$$E_k^{\text{corrected}} = T_{\text{eff}} \cdot k\, W^{k-1} \sum_{\substack{S \subseteq \{1,\ldots,8\} \\ |S|=k}} \prod_{i \in S} r_i \,. \tag{20}$$

2. **Modify the algorithm** to use a centred window $[t_0 - \tau,\, t_0 + \tau]$ around each event, accepting span up to 2τ. This sacrifices the clean greedy property (the seed would no longer be the earliest event).

3. **Leave as-is** and note the mismatch as a known conservative systematic. Given that the bias is small, well-characterised, and always in the safe direction, this is a defensible choice.